SHEET # 3

3.1 Describe the convolution and correlation operations.

3.2 Define the following:

(a) Cross-correlation between two sequences $x_1(n)$ and $x_2(n)$ having lengths N_1 and N_2 respectively.

(b) Correlation theorem.

(c) Correlation coefficients.

3.3 Use the correlation theorem to calculate the circular correlation of the data sequences {1,1,0,1} and {1,0,0,1}. Plot the correlation function against lag number, j.

3.4 Apply the correlation theorem to calculate the linear correlation of the data sequences {1,1,0,1} and {1,0,0,1}. Plot the correlation function against lag numbers and compare the results with the solution to prob.2.7, explain any differences.

3.5 Given the two sets of data {1.5, 2.0, 1.5, 2.0, 2.5} and {0, 0.33, 0.67, 1.0} calculate the following:

(a) Autocorrelation of the two sequences.

(b) Cross-correlation using direct method.

(c) Cross-correlation using correlation theorem.

3.6 Determine the output of an electrical system of impulse response function { 0, 0.899, 0.990, 0.991, 1} when the input{ 0, 2.5, 5.0, 0 } volts is applied (a) By direct convolution.

(b) By applying the convolution theorem.

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